COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	7/7/2026	**DATE SUBMITTED:**	7/8/2026
NAME:	Formilleza, Huone Anthony	**INSTRUCTOR:**	Eng. Jimlord Quejado

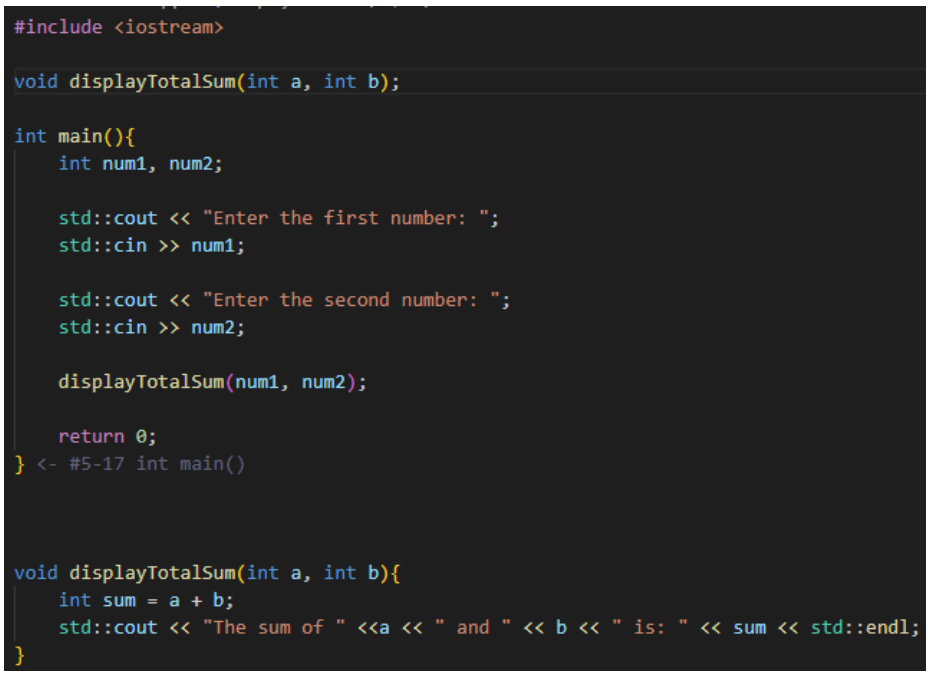
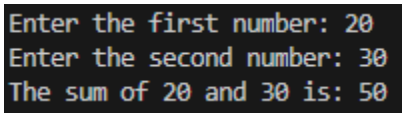
ACTIVITY NO. _2_ **[Hands-on Activity 1.1 Basic C++ Programming]**

1. PROCEDURE OUTPUT

First

Sections	**Answer**
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void displayTotalSum(int a, int b);`
Class Declaration and Method Definition Section	NONE

Main Function	`int main(){`
	`  int num1, num2;`
	
	`  std::cout << "Enter the first number: ";`
	`  std::cin >> num1;`
	
	`  std::cout << "Enter the second number: ";`
	`  std::cin >> num2;`
	
	`  displayTotalSum(num1, num2);`
	
	`  return 0;`
	`} <- #5-17 int main()`

Method Definition	
Full Code	
Output	

```
void displayTotalSum(int a, int b){
    int sum = a + b;
    std::cout << "The sum of " <<a << " and " << b << " is: " << sum << std::endl;
}
```

```
#include <iostream>

void displayTotalSum(int a, int b);

int main(){
    int num1, num2;

    std::cout << "Enter the first number: ";
    std::cin >> num1;

    std::cout << "Enter the second number: ";
    std::cin >> num2;

    displayTotalSum(num1, num2);

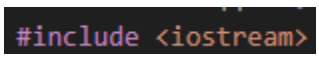
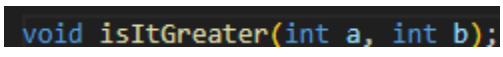
    return 0;
} <- #5-17 int main()



void displayTotalSum(int a, int b){
    int sum = a + b;
    std::cout << "The sum of " <<a << " and " << b << " is: " << sum << std::endl;
}
```

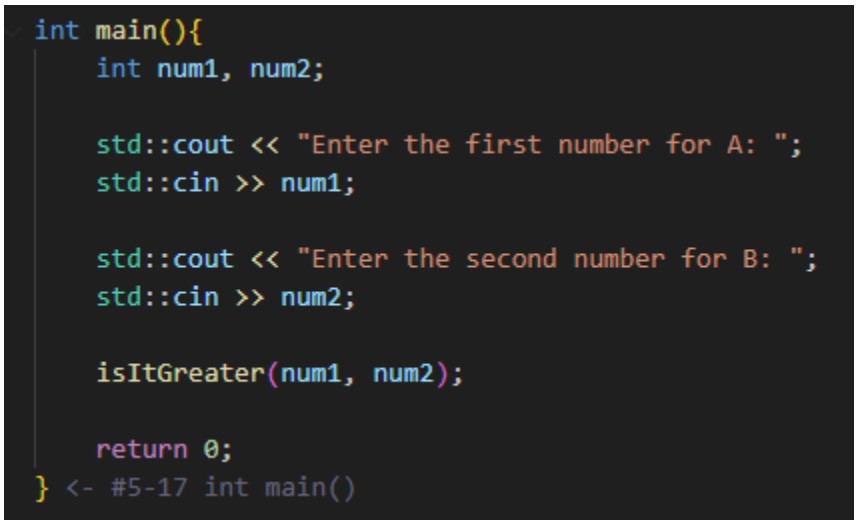
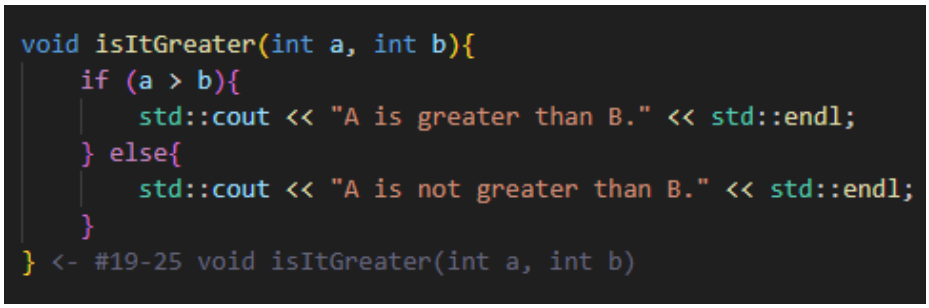
```
Enter the first number: 20
Enter the second number: 30
The sum of 20 and 30 is: 50
```

Second

Sections	**Answer**
Header File Declaration Section	
Global Declaration Section	

```
#include <iostream>
```

```
void isItGreater(int a, int b);
```

Class Declaration and Method Definition Section	NONE
Main Function	
Method Definition	

```
int main(){
    int num1, num2;

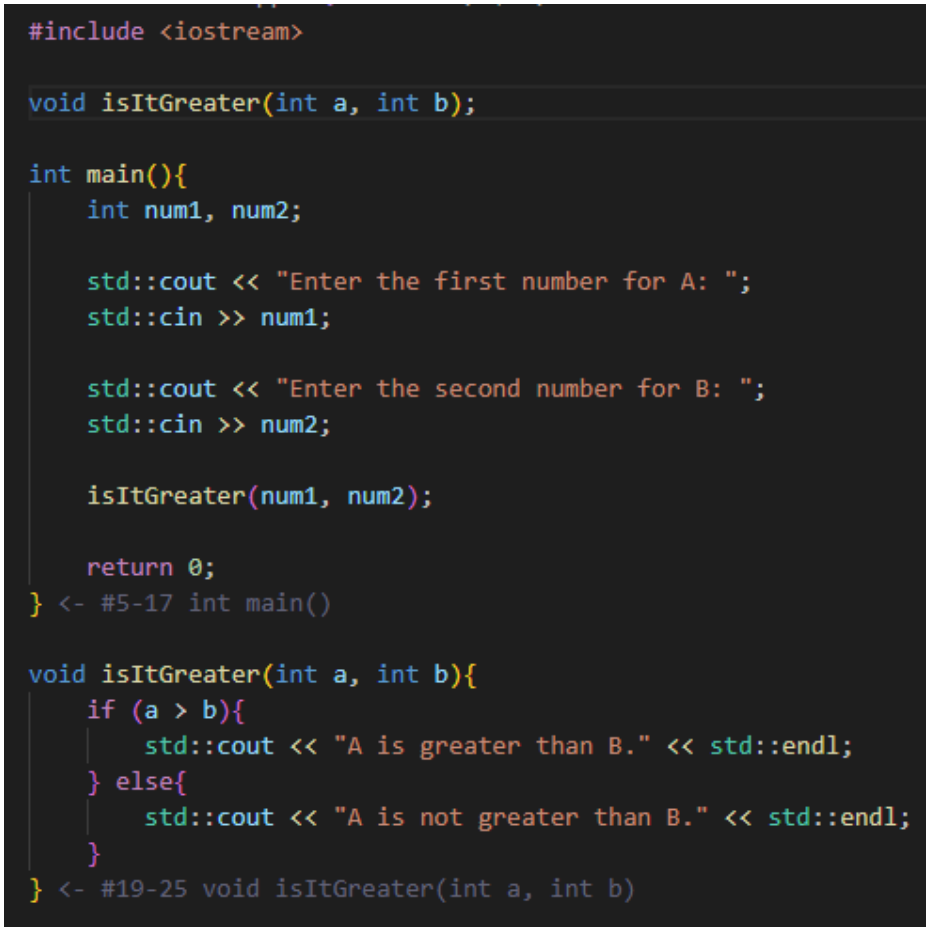
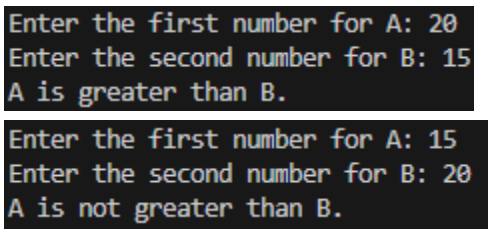
    std::cout << "Enter the first number for A: ";
    std::cin >> num1;

    std::cout << "Enter the second number for B: ";
    std::cin >> num2;

    isItGreater(num1, num2);

    return 0;
} <- #5-17 int main()
```

```
void isItGreater(int a, int b){
    if (a > b){
        std::cout << "A is greater than B." << std::endl;
    } else{
        std::cout << "A is not greater than B." << std::endl;
    }
} <- #19-25 void isItGreater(int a, int b)
```

Full Code	
Output	

```
#include <iostream>

void isItGreater(int a, int b);

int main(){
    int num1, num2;

    std::cout << "Enter the first number for A: ";
    std::cin >> num1;

    std::cout << "Enter the second number for B: ";
    std::cin >> num2;

    isItGreater(num1, num2);

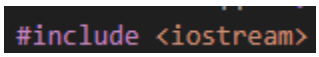
    return 0;
} <- #5-17 int main()

void isItGreater(int a, int b){
    if (a > b){
        std::cout << "A is greater than B." << std::endl;
    } else{
        std::cout << "A is not greater than B." << std::endl;
    }
} <- #19-25 void isItGreater(int a, int b)
```

```
Enter the first number for A: 20
Enter the second number for B: 15
A is greater than B.
```

```
Enter the first number for A: 15
Enter the second number for B: 20
A is not greater than B.
```

Third

Sections	**Answer**
Header File Declaration Section	

```
#include <iostream>
```

T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
COMPUTER ENGINEERING
QUEZON CITY

Global Declaration Section	

```
bool displayLogicalOperations(int a, int b);
```

Class Declaration and Method Definition Section	NONE
Main Function	

```
int main() {
    int bool1, bool2;

    std::cout << "Enter the first Boolean value (1/0): ";
    std::cin >> bool1;
    std::cout << "Enter the second Boolean value (1/0): ";
    std::cin >> bool2;
    std::cout << std::endl;
    bool isSuccess = displayLogicalOperations(bool1, bool2);
    std::cout << "\nOperation success status: " << (isSuccess ? "true" : "false") << std::endl;

    return 0;
}
```

Method Definition	

```
bool displayLogicalOperations(int a, int b) {
    std::cout << std::boolalpha;

    std::cout << "Logical Operations for " << a << " and " << b << std::endl;
    std::cout << "AND  (a && b)  : " << (a && b) << std::endl;
    std::cout << "OR   (a || b)  : " << (a || b) << std::endl;
    std::cout << "NOT  (!a)      : " << (!a) << std::endl;
    std::cout << "NOT  (!b)      : " << (!b) << std::endl;
    std::cout << "XOR  (a != b)  : " << (a != b) << std::endl;
    std::cout << "NAND !(a && b) : " << !(a && b) << std::endl;
    std::cout << "NOR  !(a || b) : " << !(a || b) << std::endl;

    return true;
}
```

COMPUTER ENGINEERING

QUEZON CITY

Full Code

HOA_1 > Third > main.cpp > displayLogicalOperations(int, int)

```cpp
#include <iostream>

// Function Prototype
bool displayLogicalOperations(int a, int b);

int main() {
    int bool1, bool2;

    std::cout << "Enter the first Boolean value (1/0): ";
    std::cin >> bool1;
    std::cout << "Enter the second Boolean value (1/0): ";
    std::cin >> bool2;
    std::cout << std::endl;
    bool isSuccess = displayLogicalOperations(bool1, bool2);
    std::cout << "\nOperation success status: " << (isSuccess ? "true" : "false") << std::endl;

    return 0;
}

bool displayLogicalOperations(int a, int b) {
    std::cout << std::boolalpha;

    std::cout << "Logical Operations for " << a << " and " << b << std::endl;
    std::cout << "AND  (a && b)  : " << (a && b) << std::endl;
    std::cout << "OR   (a || b)  : " << (a || b) << std::endl;
    std::cout << "NOT  (!a)      : " << (!a) << std::endl;
    std::cout << "NOT  (!b)      : " << (!b) << std::endl;
    std::cout << "XOR  (a != b)  : " << (a != b) << std::endl;
    std::cout << "NAND !(a && b) : " << !(a && b) << std::endl;
    std::cout << "NOR  !(a || b) : " << !(a || b) << std::endl;

    return true;
}
```

Output

```
Enter the first Boolean value (1/0): 1
Enter the second Boolean value (1/0): 1

Logical Operations for 1 and 1
AND  (a && b)  : true
OR   (a || b)  : true
NOT  (!a)      : false
NOT  (!b)      : false
XOR  (a != b)  : false
NAND !(a && b) : false
NOR  !(a || b) : false

Operation success status: true
```

```
Enter the first Boolean value (1/0): 1
Enter the second Boolean value (1/0): 0

Logical Operations for 1 and 0
AND  (a && b)  : false
OR   (a || b)  : true
NOT  (!a)      : false
NOT  (!b)      : true
XOR  (a != b)  : true
NAND !(a && b) : true
NOR  !(a || b) : false

Operation success status: true
```

```
Enter the first Boolean value (1/0): 0
Enter the second Boolean value (1/0): 1

Logical Operations for 0 and 1
AND  (a && b)  : false
OR   (a || b)  : true
NOT  (!a)      : true
NOT  (!b)      : false
XOR  (a != b)  : true
NAND !(a && b) : true
NOR  !(a || b) : false

Operation success status: true
```

```
Enter the first Boolean value (1/0): 0
Enter the second Boolean value (1/0): 0

Logical Operations for 0 and 0
AND  (a && b)  : false
OR   (a || b)  : false
NOT  (!a)      : true
NOT  (!b)      : true
XOR  (a != b)  : false
NAND !(a && b) : true
NOR  !(a || b) : true

Operation success status: true
```


2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary activity

Create a C++ program to swap the two numbers in different variables.

```cpp
#include <iostream>

void swapNumbers(int &a, int &b);

int main(){
    int num1, num2;

    std::cout << "Enter the first number: ";
    std::cin >> num1;

    std::cout << "Enter the second number: ";
    std::cin >> num2;

    std::cout << "\nBefore swapping: num1 = " << num1 << ", num2 = " << num2 << std::endl;

    swapNumbers(num1, num2);

    std::cout << "After swapping: num1 = " << num1 << ", num2 = " << num2 << std::endl;

    return 0;
}

void swapNumbers(int &a, int &b){
    int temp = a;
    a = b;
    b = temp;
}
```

Output

```
Enter the first number: 7
Enter the second number: 6

Before swapping: num1 = 7, num2 = 6
After swapping: num1 = 6, num2 = 7
```

Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Supplementary > main2.cpp > ...

```cpp
#include <iostream>

double convertKtoV(double kelvin);

int main(){
    double kelvin;

    std::cout << "Enter the temperature in Kelvin: ";
    std::cin >> kelvin;

    double fahrenheit = convertKtoV(kelvin);

    std::cout << kelvin << " K is equal to " << fahrenheit << " F" << std::endl;

    return 0;
}

double convertKtoV(double kelvin){
    return (kelvin - 273.15) * 9.0 / 5.0 + 32.0;
}

```

Output

```
Enter the temperature in Kelvin: 100
100 K is equal to -279.67 F
```

Create a C++ program that has a function that will calculate the distance between two points.

Supplementary > main3.cpp > main()

```cpp
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1, double x2, double y2);

int main(){
    double x1, y1, x2, y2;

    std::cout << "Enter the coordinates of the first point (x1): ";
    std::cin >> x1;
    std::cout << "Enter the coordinates of the first point (y1): ";
    std::cin >> y1;

    std::cout << "Enter the coordinates of the second point (x2): ";
    std::cin >> x2;
    std::cout << "Enter the coordinates of the second point (y2): ";
    std::cin >> y2;

    double distance = calculateDistance(x1, y1, x2, y2);

    std::cout << "\nThe distance between (" << x1 << ", " << y1 << ") and ("
              << x2 << ", " << y2 << ") is: " << distance << std::endl;

    return 0;
}

double calculateDistance(double x1, double y1, double x2, double y2){
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}
```

Output

```
Enter the coordinates of the first point (x1): 3
Enter the coordinates of the first point (y1): 1
Enter the coordinates of the second point (x2): -5
Enter the coordinates of the second point (y2): -1

The distance between (3, 1) and (-5, -1) is: 8.24621
```

Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

Supplementary > main4.cpp > main()

```cpp
#include <iostream>
#include <cmath>

void isItGreater(int a, int b);
int computeTrianglePerimeter(int a, int b, int c);
int computeTriangleArea(int a, int b, int c);
void classifyTriangle(int a, int b, int c);

int main(){
    int num1, num2;

    std::cout << "Enter the first number for A: ";
    std::cin >> num1;

    std::cout << "Enter the second number for B: ";
    std::cin >> num2;

    isItGreater(num1, num2);

    // New triangle part
    int a, b, c;

    std::cout << "Enter the three side lengths of the triangle (a): ";
    std::cin >> a;
    std::cout << "Enter the three side lengths of the triangle (b): ";
    std::cin >> b;
    std::cout << "Enter the three side lengths of the triangle (c): ";
    std::cin >> c;

    int perimeter = computeTrianglePerimeter(a, b, c);
    int area = computeTriangleArea(a, b, c);

    std::cout << "\nPerimeter: " << perimeter << std::endl;
    std::cout << "Area: " << area << std::endl;

    classifyTriangle(a, b, c);

    return 0;
}

```

```
40
41  void isItGreater(int a, int b){
42      if (a > b){
43          std::cout << "A is greater than B." << std::endl;
44      } else{
45          std::cout << "A is not greater than B." << std::endl;
46      }
47  }
48
49  int computeTrianglePerimeter(int a, int b, int c){
50      return a+b+c;
51  }
52
53  int computeTriangleArea(int a, int b, int c){
54      int s = computeTrianglePerimeter(a, b, c) / 2;
55      return std::sqrt(s*(s-a)*(s-b)*(s-c));
56  }
57
```

```
58  void classifyTriangle(int a, int b, int c){
59      int aSquare = a*a;
60      int bSquare = b*b;
61      int cSquare = c*c;
62
63      if (aSquare + bSquare == cSquare || aSquare + cSquare == bSquare || bSquare + cSquare == aSquare){
64          std::cout << "Triangle type: Others" << std::endl;
65      } else if (aSquare + bSquare > cSquare && aSquare + cSquare > bSquare && bSquare + cSquare > aSquare){
66          std::cout << "Triangle type: Acute-angled" << std::endl;
67      } else {
68          std::cout << "Triangle type: Obtuse-angled" << std::endl;
69      }
70  }
```

Output

```
Enter the first number for A: 10
Enter the second number for B: 20
A is not greater than B.
Enter the three side lengths of the triangle (a): 10
Enter the three side lengths of the triangle (b): 15
Enter the three side lengths of the triangle (c): 20

Perimeter: 45
Area: 60
Triangle type: Obtuse-angled
```

conclusion

The major concept involved is that functions are either self-executing (isItGreater or classifyTriangle, which prints out an output directly) or return a value back to the main for further processing (computeTriangleArea or calculateDistance). The swap function illustrated how using pass by reference helps a function to modify the values of the variables instead of using copies. The breakdown of the triangle function into perimeter calculation, area computation, and classification helped demonstrate the importance of subdividing tasks into simpler sub-tasks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini, Quillbot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **I asked gemini how to implement the the tasks in supplementary task 3 and 4 and asked what is the problem on my code and suggest a solution how to fix it, and i used quillbot to fix my grammar**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.